
Measuring accessibility: an exploration of issues and alternatives

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Abstract. Accessibility is an important characteristic of metropolitan areas and is often reflected in transportation and land-use planning goals. But the concept of accessibility has rarely been translated into performance measures by which policies are evaluated, despite a substantial literature on the concept. This paper is an attempt to bridge the gap between the academic literature and the practical application of such measures and provide a framework for the development of accessibility measures. Issues that planners must address in developing an accessibility measure are outlined, and two case studies suggestive of the range of possible approaches are presented.

1 Introduction

As congestion, crime, and housing costs have increased in metropolitan areas, many residents must question just what it is that keeps them there. The obvious answer is employment, a current job or access to future employment opportunities. The less obvious answer is access to goods, services, friends, cultural and recreational activities, and all kinds of needed and desired activities found only in metropolitan areas. In short, what keeps residents in metropolitan areas is accessibility, the potential for interaction, both social and economic, the possibility of getting from home to a multitude of destinations offering a spectrum of opportunities for work and play.

Accessibility is determined by the spatial distribution of potential destinations, the ease of reaching each destination, and the magnitude, quality, and character of the activities found there. Travel cost is central: the less time and money spent in travel, the more places that can be reached within a certain budget and the greater the accessibility. Destination choice is also crucial: the more destinations, and the greater the variety, the higher the level of accessibility. Travel choice is equally important: the wider the variety of modes for getting to a particular destination, the greater the choice and the greater the accessibility. Accessibility is thus determined both by patterns of land use and by the nature of the transportation system, although two people in the same place may evaluate their accessibility differently, as wants and tastes vary.

The term 'accessibility' has long been in the language of planners and has often made its way into generalized statements of planning goals. But the concept of accessibility defined above has rarely been translated into performance measures by which policies are evaluated and thus has had little practical impact on policies. For example, transportation planners make use of performance measures such as freeway level-of-service and average intersection delay; these measures say something about the transportation system, but in isolation, ignoring the larger context. Land-use planners focus on activity patterns, looking at characteristics such as density, and tend to ignore links between activities; traditional approaches to zoning that emphasize the segregation of land uses are a good example. The concept of accessibility, because it

accounts both for the pattern of activities and for the links between activities, provides a basis for making trade-offs between land-use and transportation policies that has been sorely lacking.

What is needed is an approach to translating the concept of accessibility into measures of accessibility that can be used to evaluate the need for and the effectiveness of alternative policies. One place accessibility measures could have a tremendous impact is in the area of the programming of transportation investments. Whether at a regional or local level, accessibility measures would provide planners and decisionmakers with a better assessment of the implications of potential investments for the daily lives of residents. Another place accessibility measures could have an important impact is in the early stages of planning, as a part of an evaluation of current land-use distributions and transportation services and in an identification of inequities and substandard conditions.

Although a substantial literature on accessibility measures has accumulated over the past four decades, what constitutes the best measure or even a good measure of accessibility is far from clear. Thus, although it is easy to say that a measure of accessibility must be developed, it is much harder to say exactly how to do this. The fundamental issue is that an accessibility measure is only appropriate as a performance measure if it is consistent with how residents perceive and evaluate their community. In other words, a practical definition of accessibility must come from the residents themselves, rather than from researchers, and reflect those elements that matter most to residents.

A framework for translating the concept of accessibility into measures of accessibility will give planners and policymakers a powerful tool for determining the need for and the effectiveness of alternative land-use and transportation policies. In this paper the beginnings of that framework are suggested, although much additional research and practical experience is needed. First, we review the literature on accessibility measures, defining several distinct but related approaches and outlining a variety of issues involved in developing an accessibility measure. Second, we present two very different case studies that suggest the range of possible approaches and demonstrate the feasibility and importance of developing measures that are easy to interpret and practical to use.

2 Alternatives and issues

The definition of accessibility stated above is based on theories of travel behavior, in particular the notion that travel is a derived demand. Mitchell and Rapkin (1954) may have first articulated this idea in stating that the amount and nature of movement derives from the amount and nature of activities. This notion remains at the heart of theories of travel behavior. Goodwin and Hensher (1978, page 25), for example, assert that travel as a derived demand implies "a simple trade-off between the advantages or benefits to be derived from being at a destination and the disadvantage or costs involved in traveling to that destination".

Similarly, most measures of accessibility consist of two parts: a transportation (or resistance or impedance) element and an activity (or motivation or attraction) element (Burns, 1979; Koenig, 1980). The transportation element reflects the ease of travel between points in space, determined by the character and quality of service provided by the transportation system and measured by travel distance, time, or cost. The activity element reflects the spatial distribution of activities. This distribution is characterized by both the amount and the location of different types of activities. The activity element is alternatively called the 'attractiveness' of a particular location as a trip destination. Others have highlighted the importance of considering a temporal

factor as well (Burns, 1979; Kitamura and Kermanshah, 1984). In fact, as discussed in section 2.2, a temporal factor is usually implicit in a measure of accessibility, as the transportation and activity elements may differ throughout the day.

Accessibility measures can be loosely organized into three types: cumulative opportunities measures, gravity-based measures, and utility-based measures. All three types incorporate both a transportation element and an activity element, although they differ in the sophistication with which they reflect travel behavior. Whatever type of measure is selected, planners must address a number of issues in developing a measure: specification, particularly the degree of disaggregation; calibration; and interpretation.

2.1 Types of measures

The simplest class of accessibility measures are the cumulative opportunities measures. These measures of accessibility count the number of opportunities reached within a given travel time (or distance). Because all potential destinations within the cutoff time are weighted equally, regardless of differences in travel time, this type of measure emphasizes the number of potential destinations or opportunities rather than their distance. This type of measure gives some sense of the range of choice available to residents, for example, in terms of the number of different stores from which they can choose. Examples of its use include McKenzie (1984), Sherman et al (1974), Wachs and Kumagai (1973), and Wickstrom (1971).

A more complex class of measures are the gravity-based measures, so called because they derive from the denominator of the gravity model for trip distribution. This type of measure weights opportunities, usually the quantity of an activity as measured by employment, by impedance, generally a function of travel time or travel cost.⁽¹⁾ Accessibility, A_i , for residents of zone i is then measured as

$$A_i = \sum_j a_j f(t_{ij}) , \quad (1)$$

where a_j is the activity in zone j , t_{ij} is travel time, distance, or cost from zone i to zone j , and $f(t_{ij})$ is an impedance function. The closer the opportunity, the more it contributes to accessibility; the larger the opportunity, the more it contributes to accessibility. Although the impedance function may take many forms, the negative exponential form has been used more often than others in recent studies and is also the most closely tied to travel behavior theory. The use of gravity-based measures has a long history (for example, Hansen, 1959; Ingram, 1971; Patton, 1976; Vickerman, 1974; Wilson, 1971).

The third, and final, class of accessibility measures is based on random utility theory, in which the probability of an individual making a particular choice depends on the utility of that choice relative to the utility of all choices. If it is assumed that an individual assigns a utility to each destination (or mode and destination) choice in some specified choice set, C , and then selects the alternative which maximizes his or her utility, accessibility can be defined as the denominator of the multinomial logit model, also known as the logsum (Ben-Akiva and Lerman, 1985; McFadden 1981). Accessibility, A_n , for an individual, n , is then measured as

$$A_n = \ln \left[\sum_{c \in C_n} \exp(V_{n(c)}) \right] , \quad (2)$$

⁽¹⁾A cumulative opportunities measure is actually a specific form of the gravity-based measure, with the impedance function equal to one if the opportunity is within the travel time limit, and zero otherwise (Koenig, 1980).

where $V_{n(c)}$ is the observable temporal and spatial transportation components of indirect utility of choice c for person n , and C_n is the choice set for person n .

The logsum serves as a summary measure indicating the desirability of the full choice set (Small, 1992). The specified utility function includes variables that represent the attributes of each choice, reflecting the attractiveness of the destination and the travel impedance that must be overcome to reach the destination, and the socioeconomic characteristics of the individual or household, reflecting individual tastes and preferences. This type of accessibility measure sometimes takes a form similar to that of gravity-based measures, but has theoretical and empirical advantages.

2.2 Specification

Regardless of the class of measure, a number of interrelated issues must be resolved in the specification of the accessibility measure: the degree and type of disaggregation, the definition of origins and destinations, and the measurement of attractiveness and travel impedance. The question of disaggregation is particularly important and has multiple dimensions. Given the spatial nature of accessibility, perhaps the most fundamental dimension is spatial disaggregation. Typically, accessibility is measured by zone, thus grouping individuals and households by proximity. The smaller the zone, the greater the disaggregation. All else being equal, smaller zones should result in more accurate estimates of accessibility for the individuals and households in the zone, as accessibility can vary greatly across small distances. Accessibility can also be measured separately for each household or individual, an approach which emphasizes the individual or the household as the decisionmaking unit. For example, Hanson and Schwab (1987) examined the underlying assumptions of accessibility measures by exploring the relationship between travel and accessibility at the individual level, and Guy (1983) analyzed the distribution of potential destinations around each household.

Accessibility measures can also be disaggregated according to socioeconomic characteristics. This dimension of disaggregation is important given that different segments of the population care about different sets of opportunities and may evaluate the impedance to and attractiveness of opportunities in unique ways. For example, Wachs and Kumagai (1973) explored the social impacts of transportation decisions and the distribution of activities on different segments of the population as defined by income and occupation categories, and Niemeier (1996) examined the worth of accessibility by income segments. In general, some differentiation of individuals and households by selected characteristics should result in more accurate accessibility measures. However, the level of disaggregation has practical limits and is likely to show diminishing returns in terms of accuracy.

The purpose of the trip or the type of opportunity represents another dimension of disaggregation. At the most aggregate level, accessibility to employment regardless of type is measured; employment serves as an indicator of overall activity. Many researchers have been concerned with evaluating the impact of transportation investments on development (for example, Hansen, 1959; Patton and Clark, 1970) or on location choice (for example, Stegman, 1969). Finer levels of disaggregation distinguish between work and nonwork opportunities. A number of researchers have been concerned with evaluating accessibility to shopping destinations and thus have included only shopping establishments in their measures (for example, Guy, 1983; Hanson and Schwab, 1987). Even more specific destination types could be singled out, depending on the purpose of the analysis, as accessibility levels to each type of destination may vary significantly for any particular zone or household.

The second issue that arises in developing accessibility measures is the origin and destination of the accessibility measure; in other words, the question of from where

and to where accessibility will be measured. Most researchers have used home-based indicators. Thus, accessibility is measured for a resident who begins or ends his or her trip at home. Given the increasing importance of non-home-based trips, the appropriateness of a home-based measure must be reevaluated (Lerman, 1979).

Multipurpose trips and trip chaining have also been highlighted as important limitations of focusing on home-based accessibility (for example, Ben-Akiva and Lerman, 1979; Richardson and Young, 1982).⁽²⁾ In a recent study, Ewing et al (1994) analyzed both home-based and non-home-based accessibility by using gravity-type measures.

The set of potential destinations to include must also be determined. The desired level of disaggregation with respect to types of opportunities is the first criterion by which destinations are screened; for example, if the intent is to measure accessibility to shopping, then only shopping destinations should be included. But the set of destination opportunities to include also depends on assumptions as to the perceived choice set, in other words, the set of potential destinations that residents perceive to be available to them (Morris et al, 1979). Researchers must ensure that the destination opportunities used in any accessibility measure reflect the needs of residents (Voges and Naude, 1983). For example, the choice sets for a study of accessibility for different socioeconomic groups should reflect the actual choices available to each socioeconomic group. Research on activity-based modeling points to the need for careful definition of choice sets and suggests that spatial and temporal constraints must be considered so that the focus is on 'constrained-choice sets' (Ben-Akiva et al, 1987; Hanson and Schwab, 1986; Jones et al, 1983). In other words, measures that include all potential destinations are likely to overestimate accessibility for any particular individual at any particular time, given his or her personal constraints and the reality of limited hours of operation for most businesses.

The measurement of travel impedance presents yet another specification issue to resolve. Distance or time, common measures of impedance, can be estimated by: straight-line distance or some modification of it (for example, Baxter and Lenzi, 1975); network models that simulate travel demand and congestion levels (Sherman et al, 1974); field surveys of actual driving times (Wickstrom, 1971); or surveys of residents with respect to perceived travel time (Wachs and Kumagai, 1973). Distance has sometimes been used instead of travel time as a measure of impedance (for example, Ingram, 1971), but the linkage to actual travel decisionmaking is less clear (for example, Koenig, 1980; Morris et al, 1979). If travel time is used, a choice must be made as to whether uncongested (or off-peak) times or congested (or peak) times will be used. The use of a generalized transport cost function, incorporating both time and monetary costs, is often an improvement over the use of time alone.

Differences in travel time and cost by mode must also be addressed. One approach is to calculate accessibility separately for different modes—automobile accessibility and transit accessibility, for example—which can then be compared. A more challenging approach is to incorporate automobile and transit travel times as well as the opportunity to travel by other modes into one measure of accessibility. Some researchers suggest that the denominator of a mode choice model, which incorporates travel times as well as other characteristics of the modes, can be used as a

⁽²⁾ However, Kitamura and Kermanshah (1984) showed that home location still largely influences the destination choice of non-home-based trips, as such stops are often made on the way home from somewhere else. Hanson (1982) compared the relative importance of home-based and non-home-based accessibility levels and found that travel patterns for employed individuals were more affected by home-based accessibility levels, though the non-home-based accessibility levels were not insignificant.

multimodal accessibility measure (Harvey and Deakin, 1993). Most current mode choice models do not reflect 'joint' mode and destination choice and thus do not incorporate variables reflecting the quality or quantity of activity at the destination. Utility-based measures facilitate multimodal measures more easily than do the other forms of accessibility measures.

The final specification issue surrounds the measurement of the attractiveness of an opportunity. This may simply be the existence of a particular opportunity, as measured by the number of establishments, or it may be either its physical or its economic size, as measured by area or employment, for example. Research on shopping behavior shows that a variety of characteristics of a potential destination are important to the destination choice (for example, Bucklin, 1967; Guy and Wrigley, 1987). This finding suggests that factors such as the quality and price of products or the quality of service, for example, could be incorporated into a measure of attractiveness. Such characteristics are highly subjective, however, making it difficult to both specify and calibrate the accessibility measure.

2.3 Calibration

Some of the specification issues can be resolved in the process of calibrating the accessibility measure; this is especially true for utility-based measures of accessibility. For all three types of measures, the goal is to calibrate the measures to reflect how individuals and households perceive the travel and destination choices available to them. Travel surveys provide a basis for calibration.

For the cumulative opportunities measures, the key element of the calibration is the choice of a cutoff travel distance or time. Accessibility levels can be highly sensitive to this cutoff. The literature does not provide a clear rule on how to make this choice. Often, a series of measures are calculated for different cutoffs (for example, Sherman et al, 1974). Frequency distributions of travel times or distances from a travel survey can provide some indication of a relevant cutoff. These types of measures are the easiest to calculate, but in many ways involve the most arbitrary calibration.

For gravity-type measures, a parameter value for the travel impedance function must be selected or estimated, by using, for example, the parameter from a calibrated trip-distribution model (for example, Voges and Naude, 1983). This parameter reflects the relative importance of travel impedance in destination choice, although for gravity models it is based on aggregate travel patterns rather than individual travel decisions. Attractiveness is often represented as simply the amount of activity, measured by numbers of employees or square footage. More complex measures of attractiveness, incorporating other characteristics of the destination, are accordingly more difficult to calibrate. Potential destinations have to be evaluated on each characteristic, by either the researcher or residents themselves, and the importance, or weighting, of each characteristic has to be determined through surveys of residents. Although usually calibrated from aggregate data, separate gravity-based measures can be calibrated for different segments of the population.

Utility-based measures are derived from the calibration of destination choice models (Basmaciyan and Schmidt, 1964). These models are calibrated from travel survey data, with each trip representing a single choice; the characteristics of the choice, including travel impedance and the attributes of both the destination and the traveler are included as explanatory variables in a utility function. The advantage of this approach is that the modeler can test alternative formulations of the utility function to find the one that best matches actual travel behavior: the relative importance (or unimportance) of different factors is determined through the calibration and does

not have to be prespecified by the modeler as it would, for example, for gravity-type measures.

Calibration is made difficult by the fact that revealed behavior—what residents actually do—is not necessarily the same as preferred behavior—what residents would choose to do given a desired set of alternatives. Revealed behavior is shaped by the specific alternatives available. For example, if a community does not have good pedestrian access, then residents will make few pedestrian trips, but this does not imply they would not make such trips if pedestrian access were better. Traditional surveys of travel behavior deal exclusively with revealed behavior, and pay little attention to the reasons behind the behavior and the underlying preferences of residents. A careful analysis of revealed behavior may hint at underlying preferences and attitudes, but new types of surveys are needed to explore travel choices fully.

2.4 Interpretation

The final issue in developing a useful accessibility measure is the process of interpreting and translating them into a form useful for policymaking. A simple approach is to compare accessibility across places, time, or both, that is, rather than focus on absolute levels of accessibility, focus on relative levels of accessibility. Relative accessibility may be defined for a particular zone or household relative to all zones or households. For example, the accessibility level of the zone can be compared with the average accessibility across all zones in the region by using a simple ratio (Patton, 1976). Changes over time may also be important, but as Dalvi and Martin (1976) stress, it is often not changes in the absolute accessibility of a zone that are important, but rather the change in the position of the zone vis-à-vis other zones on the ‘accessibility scale’. In this case, changes in relative accessibility may be the most relevant analysis.

Utility-based measures enable a more sophisticated interpretation. As Williams (1979) notes, the theory on which these measures are based provides a direct link to traditional consumer welfare theory. The model specification, that is, the calibrated utility function, can be viewed as the demand curve for a particular alternative in which a change in choice attributes, for example, a price increase, results in change in consumer surplus. The change in consumer surplus, ΔS , is simply computed as the difference in the logsums for the ‘before’ and ‘after’ scenarios:

$$\Delta S = \ln \left[\sum_{\forall c \in C_n} \exp(V_{n(c)}^A) \right] - \ln \left[\sum_{\forall c \in C_n} \exp(V_{n(c)}^B) \right], \tag{3}$$

where the observable indirect utilities before and after are indicated by a superscript B and A, respectively. The drawbacks to this measure of accessibility include interpretability and an inability to compare different utility functions, for example, by region or neighborhood. However, as case study 2 in section 4 will show, extensions in which compensating variation is used overcome both of these difficulties.

2.5 Choosing an approach

No one best approach to measuring accessibility exists; different situations and purposes demand different approaches. The trend in the transportation planning literature has been towards more disaggregate and complex representations of accessibility, at least partly in response to the recognition that aggregate measures mask many important details. As the complexity of such measures increases, however, the cost of calculation and the difficulty of interpretation increases as well. This raises two interesting questions about the choice of an accessibility measure.

First, can simple measures be as effective or perhaps even more effective than complex ones in characterizing the physical structure of a community? Although they are based rather loosely on theories of travel behavior, they may give a good, if rough, indication of accessibility. They also have the advantage of ease of understanding: cumulative opportunities measures are much easier to explain to the public than are utility-based measures or even gravity-based measures. Complexity can be added by using a series of simple measures, for example, for different types of destinations, or different measures of attractiveness. Hillman et al (1976) made use of a variety of simple measures of accessibility in their in-depth study of daily activity participation and travel choices.

Second, is it worthwhile to turn to a more qualitative approach to characterizing the accessibility of a community when a quantitative measure is deemed insufficient? Simple quantitative measures may be combined with qualitative evaluations to provide a much richer understanding of the accessibility characteristics of a community than may be possible with even very complex quantitative measures. Lynch (1981, pages 187–188) states that “while many of the obvious measures [of accessibility] are well developed, there is a gap between them and some of those felt qualities of access which citizens prize. Systematic attention to the entire range of the dimension is lacking”. More specifically, concepts of travel impedance, destination attractiveness, and choice sets can be greatly broadened to include characteristics that are not easily quantified and that depend at least partly on the perceptions of residents, rather than solely on objective criteria.

It is clear that accessibility can be measured and evaluated in a variety of ways. It is important to choose an appropriate and meaningful approach, as the use of an inappropriate measure may lead to inappropriate conclusions and ineffective policy. Different approaches may say very different things about the same community, so that an evaluation of accessibility is highly sensitive to the particular measure of accessibility used. In addition, it is not always clear how to interpret an accessibility measure and draw meaningful conclusions. The following case studies illustrate these issues of sensitivity and interpretation but demonstrate the power of creative approaches to measuring accessibility.

3 Case study 1: measuring accessibility for nonwork choices at the local scale

In the first case study we focus on measuring accessibility to nonwork activities at a local or community level. In this example, drawn from a larger study (Handy, 1992), we test the growing assumption among planners and urban designers that older, pre-World-War-2 neighborhoods have better accessibility to shops and services than have newer, postwar neighborhoods, more specifically, that they have better walking accessibility, and we find that the differences are not so clear. In this case study we demonstrate several important points about accessibility measures. First, the disaggregation of accessibility by purpose reveals substantially more than does a single measure of accessibility to nonwork activity. Second, the conclusions one can draw from the analysis of accessibility measures are highly sensitive to the specification of the measure. Third, in some cases it may be appropriate to consider separately the distance element and the activity element of accessibility. Last, an analysis of travel behavior lends support to even the simple measures of accessibility, but also highlights the limitations of developing a numeric measure of accessibility.

Two pairs of communities in the San Francisco Bay Area—each pair with one older and one newer community—were selected as case studies (table 1). These communities ranged from about 9000 population (Junior College) to over 22 000 population (Sunnyvale), and from just over one square mile (Mountain View) to nearly four

square miles (Rincon Valley); they were defined on the basis of census tracts, traffic analysis zones, and the structure of the street network. First, a gravity-based measure of local accessibility was calculated by using existing data available from the Metropolitan Transportation Commission on intrazonal travel times and amount of employment by type in each zone.⁽³⁾ Intrazonal travel times were estimated from the regional transportation planning model and are crude approximations of actual network travel times within the zone; however, they give some indication of the size of the zone from a travel standpoint. Service, retail, and other employment was used as a measure of the opportunities for nonwork activities in the zone.

Table 1. Case-study communities.

	Type ^a	Local accessibility
Silicon Valley		
Mountain View	older	2.69
Sunnyvale	newer	0.63
Santa Rosa		
Junior College	older	2.20
Rincon Valley	newer	0.49

^a Older neighborhoods are those constructed before World War 2, and newer neighborhoods are those constructed since World War 2.

This accessibility measure shows, as is often hypothesized, that the older communities have much higher levels of local accessibility than have the newer communities. Based on this accessibility measure, Mountain View and Junior College have four to six times the local accessibility of Sunnyvale and Rincon Valley. However, this measure says little about the composition and nature of those differences. It does not show whether the higher levels of accessibility are driven by shorter distances, greater amounts of activity, or both. It also does not take into account differences in the specific types of activities found there: do the high-accessibility communities contain multiple establishments of a particular type—which could very well be more than residents of the community need—and none of other types, or do they contain a full range of types, including all that residents are likely to need on a regular basis? In terms of policy direction, all that can be concluded from these measures is that Sunnyvale and Rincon Valley seem to need more commercial activity—not a particularly useful conclusion.

More detailed study of the accessibility characteristics of these communities through fieldwork tells a richer story, conveying not only the relative level of accessibility but also the nature of accessibility. Land-use patterns were first evaluated by using maps and inventories by type of establishment. In addition, commercial development was defined as being pedestrian oriented (continuous sidewalks, onstreet parking,

⁽³⁾“Local accessibility”, A_i^{local} (accessibility to commercial activity within the zone) was calculated by using the following equation:

$$A_i^{local} = \frac{E_i^{r.s.o}}{\exp(bt_{ii})},$$

where i is the origin zone, $E_i^{r.s.o}$ is the retail, service, and other employment in zone i , t_{ii} is the average intrazonal travel time, and b is a distance-decay parameter, taken to equal 0.1813. The distance-decay parameter for local accessibility was calculated by using data on convenience shopping travel from the 1980 travel diary survey of the Metropolitan Transportation Commission, 101 Eighth Street, Oakland, CA 94607.

pedestrian amenities) or automobile oriented. The distribution of land uses in each community are presented in figure 1. Mountain View and Junior College have linear strips of pedestrian-oriented commercial development as well as pockets of commercial activity within residential areas and also include strips of automobile-oriented development along major arterials. In contrast, Sunnyvale and Rincon Valley have concentrations of commercial activity in strip malls located at the intersections of major arterials. Neither newer area has any pedestrian-oriented commercial activity.

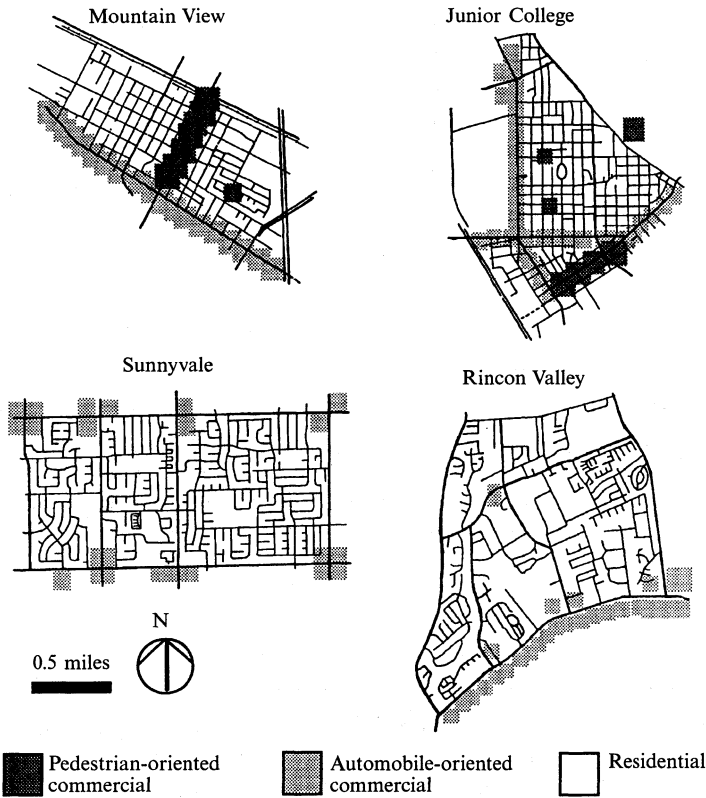


Figure 1. Street networks and land-use distributions.

Differences can be seen in the specific types of establishments found in each community (table 2). These numbers show substantial differences between these areas that were not revealed by the initial measure of local accessibility. Mountain View, for example, has no supermarket within the case-study boundaries, whereas Junior College has the greatest number on a per capita basis. With respect to other types of establishments, however, Mountain View and Junior College are more like each other than they are like the other two communities. In particular, they have several times the number of restaurants per capita and at least twice the number of convenience stores per capita. In other words, not only does the distribution and orientation of commercial activity differ between the older and newer communities, the specific types of uses vary as well. On the other hand, there is substantial variation within the older communities themselves with respect to specific types of uses.

The distribution of land uses can also be evaluated in terms of ‘coverage’, defined as the percentage of each area (but not necessarily of the population) that is within a quarter mile (about walking distance) of a supermarket, convenience store, or any

Table 2. Commercial establishments per 10 000 population.

Establishment	Mountain View ^a	Junior College ^a	Sunnyvale	Rincon Valley
Supermarkets	0.0	2.3	1.8	1.6
Convenience stores	3.7	6.8	1.3	1.6
Other food stores	4.7	10.3	4.0	2.4
Liquor stores	0.0	0.0	1.3	0.0
Restaurants	29.8	28.5	7.2	4.0
Take-out and fast food	7.5	5.7	2.7	0.8
Delicatessen or cafe	7.5	9.1	0.4	1.6
Video rentals	2.8	2.3	4.0	3.2
Cleaners	0.9	1.1	4.5	1.6
Beauty salons and barbers	6.5	16.0	4.0	2.4
Florists	5.6	3.4	0.9	1.6
Photo shops	1.9	3.4	0.9	1.6
Copy shops	2.8	6.8	0.9	0.0
Drug stores	0.0	2.3	1.8	2.4
Fitness or health club	1.9	1.1	1.3	0.8
Banks	4.7	9.1	1.8	1.6
Laundromat	0.9	2.3	0.0	0.8

^a Includes Downtown area.

Table 3. Commercial coverage: percentage of area within a quarter mile of the destination.

Destination	Mountain View	Junior College	Sunnyvale	Rincon Valley
Supermarket	0	25	17	10
Convenience store	44	59	18	13
Both a supermarket and a convenience store	0	9	6	0
Any commercial	61	90	31	20

commercial activity (table 3). Mountain View and Junior College have higher coverage than have the newer communities, but Junior College has substantially higher coverage even than Mountain View, and none of the Mountain View area is within walking distance of a supermarket. As above, these statistics highlight the fact that not all older communities were created equally. Although a very simple and easily calculated measure (particularly by means of a geographical information system), coverage provides a potentially useful evaluation of accessibility in a community.

An even better approach to this simple measure of coverage, however, is to account for network distances, rather than straight-line distances, and for the actual location of households. Distances from a sample of households (those households included in the travel survey described below) to surrounding supermarkets (among other destinations) were calculated by using the path-finding procedure of the TRANPLAN transportation modelling system.⁽⁴⁾ Then, a variety of accessibility measures were calculated based on these network distances for each of the sample households; the averages for the communities are presented in table 4 (see over).

First, by using a cumulative opportunities formulation, the number of supermarkets within particular travel times from home were counted for the sample households. One thing about this measure is especially notable: the differences between communities depend on the travel time selected. Second, the automobile travel time

⁽⁴⁾ Available through Urban Analysis Group, 50 Oak Court, Suite 110, Danville, CA 94526.

Table 4. Measures of supermarket accessibility.

	Mountain View	Junior College	Sunnyvale	Rincon Valley
Average number of supermarkets within:				
2 minutes	0.1	0.2	0.3	0.2
5 minutes	5.5	3.8	5.0	3.8
10 minutes	7.0	9.0	9.0	7.8
Minimum time to supermarket (minutes)	2.9	2.4	2.5	2.8
Time-weighted number of supermarkets	0.9	0.8	1.0	0.7

Note: in each case the *F*-statistic indicates greater variation than within case-study areas.

to the nearest supermarket was determined. This represents the minimum time that it takes a resident to reach a supermarket. Although this measure does not account for the variety available to residents, it does reflect the most basic level of accessibility. If variety matters to residents, then this measure is only a partial indication of accessibility; if distance is the sole concern of residents, then this measure is a relatively complete indication of accessibility.

Third, a gravity-based measure of supermarket accessibility was calculated, incorporating the travel time to all supermarkets within a chosen limit.⁽⁵⁾ The farther away a supermarket, the more it was discounted, and the greater the time-discounted number of supermarkets, the better the accessibility. The advantage of this measure is that it reflects both travel time and variety of potential destinations. The disadvantage, as with the aggregate measure of local accessibility calculated above, is that it is not so easy to interpret. However, this measure does provide a more specific evaluation: note that supermarket accessibility is much more equal across the case-study areas than was the general measure of local accessibility. This finding points to the importance of disaggregating accessibility measures by activity; more aggregate measures may mask important differences.

One advantage of calculating accessibility for households, rather than for a community as a whole, is that it is possible to evaluate the distribution of accessibility within the community, in other words, whether households have relatively equal accessibility or whether some households have much better accessibility than others. The distribution of supermarket accessibility, measured as the time-discounted number of supermarkets, shows significant variation around the mean level of accessibility in each case-study area (figure 2). Sunnyvale had the highest average for the sample of households and also had the flattest distribution; in other words, supermarket accessibility in this area varies more across households than it did in the other case-study areas.

⁽⁵⁾ The time-discounted number of supermarkets (*N*) for each household was calculated by using the following equation:

$$N_i = \sum_j \exp(-bt_{ij}),$$

where *i* is the origin household, *j* is the destination supermarket, *t_{ij}* is the travel time from household *i* to supermarket *j*, and *b* is the distance-decay parameter, taken to be equal to 0.52. The distance-decay parameter was calculated by using data from the neighborhood travel survey.

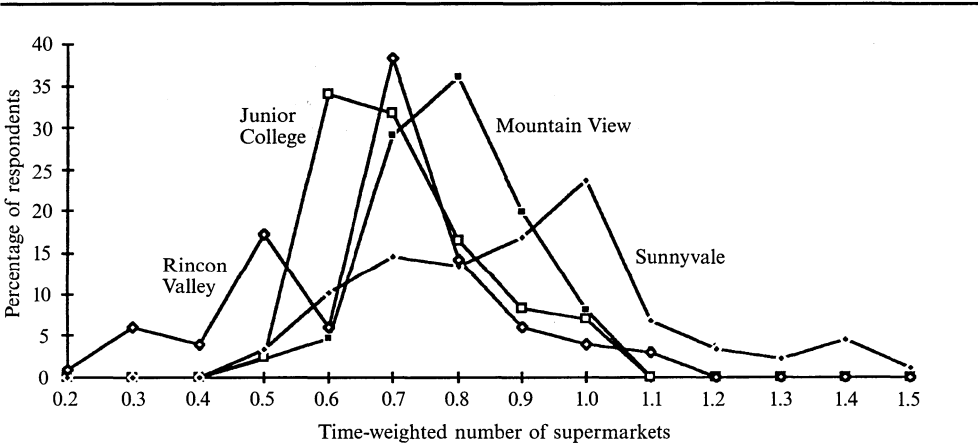


Figure 2. Distribution of supermarket accessibility within the case-study areas.

The appropriateness of these measures can be tested through an analysis of travel behavior. As a part of the larger research effort (Handy, 1992), residents in each of the case-study communities were surveyed with respect to various characteristics of their nonwork travel patterns. One hundred residents in each case-study area were surveyed by phone over a one-week period in April 1992. Although most of the focus was on revealed travel behavior, some attempt was made to identify the reasons for particular travel choices. In the following examples we look at the importance of a range of choice, which is central to the concept of accessibility, and the characteristics of destinations, which many researchers have suggested should be defined more broadly than simply by quantity of activity.

In order to evaluate the importance of having a variety of destinations from which to choose, respondents were asked both about the frequency of visits to different supermarkets and about their usual choice of supermarket. That variety is important to residents is demonstrated by their use of more than one supermarket on a regular basis. First, most residents visit more than one supermarket at some time. Only 17% of respondents visited only one supermarket in a month, whereas 41% visited three or more; on average, residents visited 2.45 different supermarkets over a month (table 5). Second, not all residents in an area named the same supermarket as their usual choice: seven or more different supermarkets were named as the usual choice in each of the case-study communities.

Table 5. Variety in supermarket choice.

Supermarkets visited ^a	Respondents		Cumulative percentage
	number	%	
1	67	17	17
2	168	42	59
3	97	24	83
4	50	13	96
5	15	4	99
6	3	1	100
Total	400		
Average number	2.45		

^a In one month.

Although traditional measures of accessibility focus on the quantity of activity and the cost of travel, it is clear that many other factors affect destination choice. When asked why they frequented a particular supermarket, about as many respondents indicated that product quality and the range of product selection were major reasons as those who indicated distance (table 6). Atmosphere was a major reason for just over half of the respondents, and prices were a major reason for just under half. These findings suggest that distance and quantity are important, but so are quality and variety. In other words, destination attractiveness needs a broader definition than is provided for in most measures of accessibility.

Table 6. Reasons for choice of usual supermarket.

Reason	Percentage of respondents		
	major reason	minor reason	not a reason
Product quality	66.5	24.8	8.7
Distance	65.5	22.3	12.2
Widest selection	64.8	26.1	8.9
Atmosphere	52.9	33.5	13.6
Prices	48.9	39.5	11.7
Parking	37.5	36.2	26.3
Stores nearby	24.8	32.5	42.7

4 Case study 2: measuring accessibility for work mode and destination choices at the regional scale

The second case study, also drawn from a larger study (Niemeier, 1996), illustrates an innovative approach to interpreting a utility-based measure of accessibility in the context of evaluating alternative scenarios. This approach is based on the concept of compensating variation, which may be interpreted as the dollar amount an individual would have to be compensated by to be as well off as before a policy change (Freeman, 1993; Zerbe and Dively, 1994). By using this concept, we can translate changes in accessibility under different scenarios into monetary terms that can be easily compared and interpreted (Small and Rosen, 1981); these monetary values represent the ‘worth’ of accessibility under a particular scenario relative to a base scenario. In addition, accessibility measures calculated by income group are used to evaluate the differential worth of accessibility for population segments. Although theoretically complex, the method is not difficult to apply and generates an elegant evaluation tool.

The method was applied to King County, Washington. Accessibility in this example is measured for work destinations and the morning peak period of travel. Three primary sources of data were used: waves 1 and 2 of the Puget Sound Regional Transportation Panel travel survey data, the Puget Sound Transportation Model, and the 1990 census data (PSRC, 1990). The example begins with the specification of an indirect, or observable, utility function describing the desirability of a specific choice of mode and destination for the journey to work. The indirect utility is specified as a function of attributes which vary across modes and destinations and variables reflecting individual tastes and preferences:

$$V_{n(c)} = f\left(\frac{e_{n(c)}}{Y_n}, Z_n, T_{n(c)}, D_{n(c)}, \varepsilon_n\right), \tag{4}$$

where $e_{n(c)}$ is the cost of travel for mode-destination combination c for person n , Y_n is the household income for person n , Z_n are the socioeconomic characteristics for person n , $T_{n(c)}$ are the transportation choice attributes for person n for choice c , $D_{n(c)}$ are

the destination choice attributes for person n for choice c , and ε_n is the unobserved portion of utility for person n .

A wide range of variables were tested in the specification of the utility function, with the final model including the ratio of modal travel cost to household income, indicator variables identifying women traveling by transit and elderly traveling to the central business district (CBD), variables capturing attributes of modal travel time, and variables indicating the proportion of jobs, by type, in the destination analysis zone:

$$V_{n(c)} = \beta_{0c} + \beta_1 \left(\frac{e_{n(c)}}{Y_n} \right) + \beta_{2n} Z_n + \beta_{3c} T_n + \beta_{4c} D_n , \tag{5}$$

where the β are coefficients estimable by standard multinomial logit methods (Train, 1986).

A variety of policy scenarios were then constructed. For example, a mode is removed from the mode–destination choice set. The difference in logsums then reflects the contribution to accessibility of the removed alternative. However, as previously noted, accessibility in this form is difficult to interpret and compare. Thus, an additional computation, facilitating comparability, is desired.

The multinomial logit form of accessibility can be translated into monetary terms by using compensating variation, v (Small and Rosen, 1981). The difference in logsums, Δv_n , is computed by using equation (3) and then multiplying by the inverse of the marginal utility of income, λ :⁽⁶⁾

$$\Delta v_n = -\frac{1}{\lambda} \left[\ln \sum \exp V_{n(c)} \right]_{V_{n(c)}^A}^{V_{n(c)}^B} . \tag{6}$$

Thus, the compensating variation values represent the accessibility worth or the amount someone must be compensated by after a policy change that reduces accessibility in order to be as satisfied as before the policy change.

Alternative scenarios are constructed by removing a particular mode or destination (or combination of mode and destination) from the choice set and comparing with the existing, or base case, choice set. The methodology is applicable to scenarios in which the utilities of particular choices change as a result of, for example, a decrease in travel time for a mode or an increase in employment at a certain destination.

The construction of simple scenarios in which a mode or destination is removed yields some interesting results. For example, if the option to drive were eliminated, an individual, on average, would require a minimum compensation of \$5.30 per trip to be as well off as before the policy change, as shown in table 7 (see over). In contrast, the elimination of transit results in an average consumer welfare loss (that is, requires a compensation) of \$0.87 per morning journey-to-work trip. Thus, the compensation required to make an individual indifferent to eliminating the auto option is six times as high as for the transit option.

Likewise, the change in consumer welfare may also be computed when accessibility is reduced by limiting the destination choices. In this policy scenario, the compensating

⁽⁶⁾If a price of cost variable is included in the indirect utility specification, λ can be computed by using Roy's identity (Small and Rosen, 1981):

$$\lambda_g = -\frac{1}{x_g} \frac{\partial V_g}{\partial C_g} ,$$

where x_g is the individual's consumption of good g . For a model specification in which the ratio of cost to income is included, λ is equal to the cost to income ratio coefficient in equation (5) divided by the household income Y_n .

variation represents the minimum compensation required to make an individual, on average, indifferent to the loss of accessibility to selected subarea destinations (consisting of clusters of census tracts) having greater than average job opportunities. For example, as shown in table 7, an individual, on average, would require a minimum compensation of \$2.23 to retain the same level of accessibility worth if the CBD were eliminated as a potential work destination. Further, accessibility to the CBD is worth, on average, at least 4.2 times more than accessibility to the south area (\$0.53) and 11.2 times more than accessibility to the eastside area (\$0.20).

Table 7. Compensating variation for mode and subarea policy scenarios.

Policy scenario	Compensating variation (\$)
Mode eliminated:	
auto	5.30
transit	0.87
Subarea destination eliminated:	
central business district	2.23
south	0.53
eastside	0.20

This approach can also be used to reveal important differences between income groups. As shown in table 8, individuals with household incomes below \$35 000 would require slightly more than 1.5 times the compensation for the loss of transit accessibility than those with incomes greater than \$35 000.⁽⁷⁾ Individuals in the higher income market segment would require nearly 5 times the compensation of individuals in the lower market segment after a loss in auto accessibility to be as well off as before the policy change. Of particular interest is the ratio between auto and transit compensating variations. Individuals with household incomes less than \$35 000 require approximately the *same* amount of compensation to be as well off as before policies which restrict the choice to either auto or transit. This speaks to the relative importance of transit accessibility for the lower income segments.

We also present in table 8 the compensating variation for income segments for the elimination of the subarea destinations. The results suggest that elimination of the CBD as potential work destination requires a compensation of \$1.88 and \$2.32 for

Table 8. Compensating variation by market segment for mode and subarea policy scenarios.

Policy scenario	Compensating variation (\$)	
	income <\$35 000	income >\$35 000
Mode eliminated:		
auto	1.26	6.33
transit	1.29	0.77
Subarea destination eliminated:		
central business district	1.88	2.32
south	0.26	0.60
eastside	0.08	0.23
Mode to central business district eliminated:		
auto	0.35	1.45
transit	1.11	0.55

⁽⁷⁾ The median income for the State of Washington is US\$31 181, whereas the median incomes for King County and Puget Sound are US\$36 179 and US\$35 161, respectively. Thus, US\$35 000 represents a useful boundary for examining broad market segments.

households earning less than \$35 000 and more than \$35 000, respectively, to maintain the same level of welfare as before the change. That the compensation is only about 20% more for individuals with household incomes over \$35 000 than for individuals with household incomes below \$35 000 suggests that CBD accessibility is similarly valued by both market segments.

A greater difference is found when comparing compensation variations to the eastside or south subareas, both more suburban in nature. On average, individuals having household incomes greater than \$35 000 would require roughly three times the compensation of individuals with household incomes below \$35 000, by any mode, if accessibility to either the east or south subareas were eliminated.

These findings highlight the importance of transit accessibility to the CBD for the lower income segment; the compensating value of transit accessibility to the CBD for individuals with household incomes under \$35 000 is three times higher than that of auto accessibility, reflecting a generally greater reliance and value placed on this mode. In contrast, for individuals with household incomes over \$35 000, the compensating value of auto accessibility to the CBD is almost three times higher than that of transit.

5 Conclusions

In the USA, State Departments of Transportation and Metropolitan Planning Organizations are increasingly turning to accessibility measures in their planning processes, although they are struggling with how to measure accessibility practically. Rutherford (1994) provides an extensive review of the types of accessibility measures being considered in infrastructure planning analyses. For the most part, these measures focus on the transportation component of accessibility in terms of travel times and speeds; they lack a sufficient consideration of the activity or destination component to be adequate measures of accessibility (table 9, see over). The Western Bypass Study, in Portland, Oregon, is something of an exception. In this study accessibility was measured by the percentage of the study area within 30 minutes of 800 000 population, within 30 minutes of 500 000 jobs, and within 15 minutes of 25 000 retail jobs—a measure similar to the measure of coverage in case study 1.

Other examples also demonstrate the ongoing struggles to define measures of accessibility. Santa Clara County, California, developed a set of performance measures including accessibility and other considerations to be used in the development of its state-mandated Congestion Management Program (Cambridge Systematics, Inc., 1995). Accessibility is measured at the zonal level as the percentage of the employed population in a zone within 20 minutes of employment activity. Separate measures are calculated by using travel times for four different modes, and these measures are then factored by the percentage of households within the zone having access to that mode. Montgomery County, Maryland, uses a gravity-type measure of accessibility, based on travel times to and the amount of employment in the destination zone. Separate measures are calculated for each mode (Levinson and Kumar, 1994). An overall measure of accessibility for the county is defined as the average of accessibility for all zones, weighted by the number of households in each zone. Both counties, however, are just beginning to determine how best to interpret these measures and incorporate them into their planning processes.

We have outlined important issues that transportation agencies must address in choosing an accessibility measure and have presented two very different case studies suggesting several possible approaches. In the first case study we focused on accessibility to shopping activities at the local scale. In this example we used simple but relatively disaggregate measures of accessibility. These measures made use of data

Table 9. Multimodal evaluation measures in practice (source: Rutherford, 1994).

Location	Study type	Accessibility measure
Sacramento–San Francisco corridor study	Intercity	Afternoon peak-hour travel time between selected destinations
Maryland Statewide Commuter Assistance Study	Intercity	Travel time between selected destinations Increased access to areas of economic development
Hali 2000 regional study	Regional	Travel time to major employment centers
Vision 2020, Seattle, WA	Regional	Level of public service
HOV corridor screening, Chicago, IL	Regional	Travel time savings—afternoon peak period
I-80 HOV study, New Jersey	Corridor	Travel time savings for HOV users
I-40 HOV feasibility study, Raleigh, NC	Corridor	Travel speed Travel time between key points
I-15/State Street corridor, Salt Lake City, UT	Corridor	Selected auto travel times Selected transit travel times
Western Bypass Study, Portland, OR	Corridor	Percentage of study area within 30 minutes of 800 000 population Percentage of study area within 30 minutes of 500 000 jobs Percentage of study area within 15 minutes of 25 000 retail jobs

Note: HOV, high occupancy vehicle.

that are not readily available but that are relatively easy to collect. In the second case study we focused on accessibility by different modes to work destinations at the regional scale. In this example we used a utility-based measure of accessibility and an innovative approach to interpreting and evaluating the results in monetary terms. Although the theory behind this approach is complex, it can be applied with travel-diary data that are readily available (although sometimes dated) in most large metropolitan areas. In both case-study approaches we incorporated spatial and temporal elements in the respective measures of accessibility.

A distinct gap currently exists between the academic literature and the practical application of accessibility measures. It is important that accessibility measures used in practice are theoretically and behaviorally sound and that innovative approaches to measuring accessibility are made practical. In this paper we have identified several practical approaches to measuring accessibility which seem uniquely suited for moving beyond the constraints of current aggregate modeling approaches and incorporating recent findings in activity-based travel-behavior research, in which the importance of considering endogenous trip-scheduling, trip-chaining, and nonwork travel has clearly been demonstrated. As funding resources for transportation infrastructure become increasingly constrained and as communities turn to land-use policies, such as transit-oriented development, to manage demand, the use of accessibility measures reflecting person-scale dimensions and concerns will be critical in ensuring that infrastructure improvements and land-use policies serve and speak to individual activity needs.

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